

Swarnima Jaiswal

+91 9335748363 | jaiswalswarnima1@gmail.com | GitHub | LinkedIn

Summary

AI/ML-focused B.Tech student with hands-on experience in Machine Learning, Computer Vision, Generative AI, and Web Development. Built AI-driven projects including a CNN-based Face Mask Detection system and a Financial RAG Chatbot, along with industry experience as a Web Development Intern working with React.js, Node.js, and Next.js. Strong foundation in programming and eager to apply AI/ML and software development skills to real-world problems.

Education

Vellore Institute Of Technology, Bhopal

2023 – 2027

B.Tech in Computer Science and Engineering (Artificial Intelligence and Machine Learning), **CGPA: 8.60/10**

Technical Skills

Programming Languages: Python, SQL, Java

AI & Machine Learning: Artificial Intelligence, Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Computer Vision

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, LangChain, NumPy, Pandas, Streamlit

Backend & APIs: FastAPI

Developer Tools: GitHub, VS Code, PyCharm, Jupyter Notebook, Google Colab

Core Computer Science: Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Database Management Systems (DBMS)

Experience

EdTech Innovate Pvt Ltd

May 2026 – Jul 2026

Web Development Intern

Noida, UP

- Developed responsive web pages and implemented dynamic features using **React.js**, focusing on UI optimization and user experience.
- Worked with **Node.js** for backend development, including APIs and server-side logic to ensure smooth data flow.
- Explored **Next.js** for modern, scalable web applications and contributed to debugging, testing, and application maintenance.

Projects

Financial RAG Chatbot | Python, QLoRA, RAG, Hugging Face, Vector Search

July 2026

- Built a **Retrieval-Augmented Generation (RAG) chatbot** for answering finance-related queries from documents.
- Worked with a QLoRA + RAG-based financial question-answering system using **Qwen2.5-1.5B-Instruct**, focusing on understanding fine-tuning and retrieval-augmented generation workflows.
- Explored the complete RAG pipeline including **data preparation, document chunking, embeddings, vector retrieval, prompt construction, and LLM-based response generation**.
- Ran and analyzed experiments on **FinanceBench**, comparing base, fine-tuned, and RAG-based approaches and studying the impact of retrieval and strict numeric prompting on financial QA performance.

AI Interview Assistant | Python, FastAPI, React.js, Vite, Groq, GPT-OSS-20B, PyMuPDF

May 2026

- Built a full-stack AI-powered **AI/ML interview platform** that analyzes candidate resumes and generates personalized technical interview questions based on their skills, projects, and experience.
- Developed **FastAPI REST APIs** for PDF resume parsing, interview session management, question generation, answer submission, and AI-based evaluation using Groq-hosted LLM inference.
- Implemented **adaptive interviewing** where subsequent questions are generated based on the candidate's previous answer, evaluation, resume context, and previously asked questions.
- Designed an automated evaluation system scoring answers on **technical accuracy, clarity, depth, and relevance** with strengths, weaknesses, feedback, and question-wise scores.
- Built a final **performance analytics dashboard** displaying total score, average score, category-wise performance, and question-wise results across the complete interview.

Certifications

- Introduction to Machine Learning - Great Learning
- Applied Machine Learning in Python - Coursera
- Google IT Professional Certificate - Google